



THE ROMAN CATHOLIC DIOCESE OF SUMBAWANGA

**FORM FOUR MOCK EXAMINATION FOR
DIOCESAN SECONDARY SCHOOLS 2026**



031/01

PHYSICS I

Time 3:00 Hours

2026.

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and only **two (2)** questions from section C.
3. Communication devices and any unauthorized materials are **not** allowed in the examination room.
4. **Non-programmable** calculators and mathematical tables may be used.
5. Write your **Examination Number** on every page of your answer booklet (s)
6. Where necessary the following constants may be used;
 - (i) Acceleration due to gravity, $g = 10\text{m/s}^2$ or 10N/kg
 - (ii) Density of water = 1g/cm^3
 - (iii) Density of xylene = 880kg/m^3
 - (iv) Linear expansivity of copper = $0.000017^\circ\text{C}^{-1}$

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SECTION A (16 Marks)

Answer all questions

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet provided.

- (i). When a force is applied to a given material, the material returns to its original shape after the force is removed. What is the name given to this behaviour?
A. Plastic deformation B. Elastic deformation C. Viscous deformation
D. Irreversible deformation D. Neutral deformation
- (ii). A uniform beam XY is pivoted at one end X, and vertically upward force of 4N at Y keeps the beam horizontal. From the information given above, what is the weight of the beam in Newtons?
A. 8 B. 6 C. 4 D. 2 E. 1
- (iii). Which of the following statements is not the reason why a microscope has an objective lens of short focal length?
A. It allows more light to be collected B. It makes the image formed to be inverted
C. It keeps the distance between the objective lens and eyepiece lens small.
D. It gives the microscope more magnifying power.
E. It gives the microscope coloured images
- (iv). A car moving at steady speed has a frictional force on its surface. What is the correct factor in which the frictional force depends?
A. Its speed and surface area B. Its speed only C. Its surface area only
D. Its weight E. Its wheel speed only
- (v). The kelvin temperature of a liquid is 300K. What is its temperature in °C ?
A. 273 B. 100 C. 57 D. 37 E. 27
- (vi). A commercial aeroplane is cruising at an altitude above sea level. At take off the plane was at a pressure of 1 atm. What additional pressure should the cabin be pressurized to normalize the pressure to 1 atm at this height if a pressure measuring instrument indicates a height difference of 600 mmHg?
A. 70000 dm³ B. 10 dm³ C. 10000 dm³ D. 100 dm³ E. 1 dm³
- (vii). Illuminated smoke particles, suspended in air, are viewed with a microscope and are seen to vibrate continuously. What can you conclude about the observation on the smoke particles?
A. They are bombarded by continually by air particles
B. They are supplied with energy by the light illuminating them
C. They are agitated by sound waves from the surrounding room
D. They are shaken by the vibration of the molecules within them
E. They are moved around by convection currents
- (viii). An aluminium cube has a volume of 800 cm³. If it is totally immersed in water. Assume density of water 1000kg/m³ and acceleration due gravity is 10N/kg. What will be the upthrust acting on it?
A. 8N B. 0.8N C. 80N D. 10N E. 18N
- (ix). In physics test students were asked to mention the examples of friction. The response of one candidate was as listed below.
1. Tension in a spring.
2. Air resistance.
3. Weight.

Which among the alternatives shows the correct examples of friction?

- A. 1 only B. 1 and 3 C. 2 only D. 2 and 3 E. 1 and 3

- (X) When a capacitor is charged and disconnected from a battery, what will happen to its charge and voltage if the distance between the plates is increased?
- The charge will remain constant, but the voltage will increase.
 - Both charge and voltage will remain constant.
 - The charge will decrease, but the voltage will remain constant.
 - The charge will remain constant, but the voltage will decrease.
 - Both charge and voltage will be varied proportionally.

2. Match the concepts of magnetization from **List A** with the appropriate terms in **List B** by writing the letter of the correct response beside the item number in answer sheet provided.

List A	List B
(i) It is the process of aligning the atomic dipoles in a material in one direction so as to produce a net effect of attraction or repulsion.	A. Double touch
(ii) Materials in which it is possible to cause this alignment of magnetic domain easily.	B. Saturation
(iii) Material such as aluminium and chromium.	C. Single touch
(iv) A portion inside a magnetic material where the atomic dipoles align in the same direction.	D. Magnetic domain
(v) The stroking method whereby only one magnetizing magnet is used.	E. Magnetic dipole
(vi) A state when no further alignment of magnetic domains takes place.	F. Paramagnetic materials
	G. Ferromagnetic materials
	H. Magnetization
	I. Demagnetisation
	J. Unsaturation

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) Suppose you are helping in designing a magnetic tool for lifting nails in a workshop. However, the magnet gets demagnetized frequently due to rough handling. Propose an improved design or method that ensures the tool stays magnetized for a longer time, and explain why your design would work better. **(5 marks)**
- (b) Suppose you are in physics laboratory and you are given a bar magnet whose poles are not labelled. Describe how you could determine which end of the bar is the North Pole. **(4 marks)**
4. (a) Why do small droplets of pure water evaporate in air even the relative humidity is 100%? **(4 marks)**
- (b) A room with dimensions of 7m x 10m x 2m, holds air that is saturated with water vapour. The saturation vapour pressure of the water vapour is 7.37kPa. If all of the water vapour. **(5 marks)**
5. (a) Students are performing two experiments. One is on latent heat of vaporisation and other is on latent heat of fusion. In which to use a large amount of energy against the environment? **(3 marks)**
- (b) An empty barometer tube, 1m long, is lowered vertically downwards, into a tank of water. What will be the depth at the top of the tube when the water has risen 20 cm inside the tube? (Atmospheric pressure may be assumed equal to 10.4m column of water). **(6 marks)**
6. (a) A small coin was placed at the bottom of a tall glass containing some water and viewed from above. The real and apparent depths of the coin were then measured. By varying the depth of the water in the jar, the following readings were obtained: **(6 marks)**

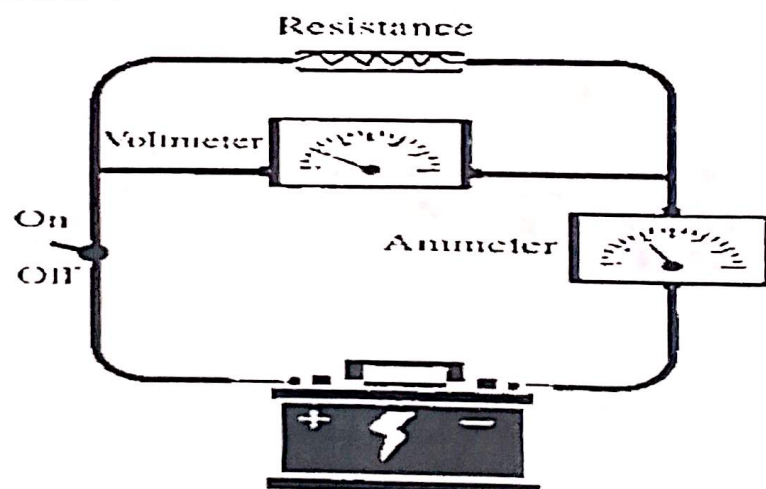
Real depth (cm)	8.1	2.0	16.0	20.0
Apparent depth (cm)	5.9	9.0	12.0	15.1

- By plotting an appropriate graph from the results, determine the refractive index of water.
- (b) Using a ray diagram, show that if an object is placed at the focal point of a concave mirror, the image will be formed at infinity. (3 marks)
7. (a) The distance from the cornea to the retina of someone's eye is 2.1 cm. Determine the effective focal length of this eye of an object is at infinity, located 1 m away and located 25 cm away respectively. (6 marks)
- (b) An astronomer wants to order a large concave mirror for a telescope that is to produce high quality images. With explanation, advise whether the astronomer should order a spherical mirror or a parabolic mirror? (3 marks)
8. (b) With the aid of a diagram, explain when Venus is termed as a morning star and evening star. (5 marks)
- (b) Write and describe briefly any two electromagnetic radiation types that can be observed by a space-based telescope and two that can be observed by a ground-based telescope. (4 marks)

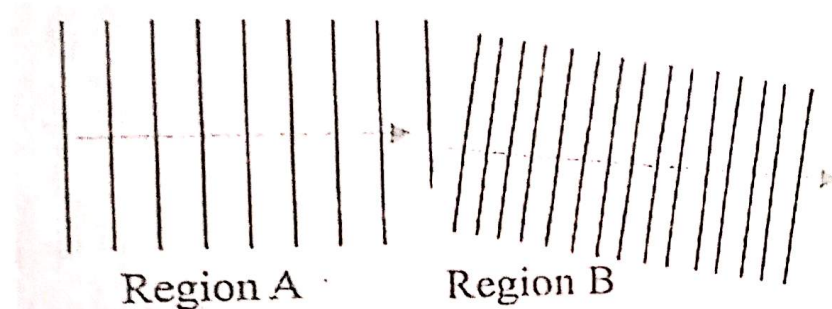
SECTION C (30 Marks)

Answer any two (2) questions

9. A student set up the schematic diagram as in Figure below to investigate the internal resistance of a cell where $R = 2\Omega$ is a fixed resistor, r is the internal resistance of the cell, A is an ammeter and V is a voltmeter across the cell. When the switch is closed, the ammeter reads 0.4 A, the voltmeter reads 1.2 V. If the emf, E of the cell is known to be V .



- (a) Draw the circuit diagram clearly using circuit symbol and label all components. (7 marks)
- (b) Calculate the effective resistance of the entire circuit. (4 marks)
- (c) Use the readings to calculate the internal resistance r of the cell. (4 marks)
10. (a) Figure below shows water waves in a ripple tank moving from one to another region. The two regions have two depths.



- (i) In which region are the waves moving faster? Explain. (4 marks)
- (ii) Explain which region of the ripple tank is deeper. (4 marks)
- (b) In a resonance tube experiment, the smallest value of l for which a peak in sound intensity occurs is 9.0 cm. How far the tube must be raised to hear the next peak in sound intensity. Neglect the end correction. (7 marks)
11. (a) In an experiment where an electric current flows through a conductor, a clockwise magnetic field is produced around the conductor. If a teacher decides to reverse the terminals of the battery, what would be the effect on the magnetic field? (5 marks)
- (b) The element tin has a total of twenty-five isotopes with the latest isotope being $^{108}_{50}\text{Sn}$. Assuming that all twenty-five isotopes exist and the difference in mass numbers of two successive isotopes is one, write down the symbol for the heaviest isotope. (6 marks)
- (c) For the conclusion given below, state its proof from the Rutherford's experiment. (4 marks)
- "Most of the mass of the atom is concentrated in a tiny region called the nucleus".*